

Existence of Non-Equivariant Gromov Tori

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Result

Theorem (R. Bryant, 1985). $\mathbb{T}^2$ is a flat torus isometrically and minimally embedded in $\mathbb{S}^N$. Then it is equivariant.

Theorem 1. $n \geq 3$. Then $\exists \mathbb{T}^n$, a flat torus isometrically and minimally embedded in $\mathbb{S}^N$, which is not equivariant.

Theorem (M. Gromov, 2022). $n \geq 2, N \gg n$. Then

$\exists \mathbb{T}^n \subset \mathbb{T}_{\text{Cl}}^N \subset \mathbb{S}^{2N-1}$, where $\mathbb{T}^n$ is a geodesic subtorus, $k_p(v) \equiv \sqrt{\frac{3n}{n+2}}$ for $p \in \mathbb{T}$, where $k_p(v)$ is the normal curvature. Also, $\mathbb{T}^n$ is equivariant.

Corollary (M. Gromov, 2022). (L, g) is a Riemannian manifold. Then $\exists \lambda > 0 : (L, \lambda^2 g) \hookrightarrow \mathbb{B}^q$, $k_p(v) \leq \sqrt{3}$ for $p \in L$.

Theorem (A. Petrunin, 2023). $\mathbb{T}^n \looparrowright \mathbb{B}^q$. Then
$$\max\{k_p(v) \mid p \in \mathbb{T}, v \in T_p \mathbb{T}\} \geq \sqrt{\frac{3n}{n+2}}.$$

Result

Definition. Let $\mathbb{T}^n \looparrowright \mathbb{B}^q$ and $k_p(v) \leq \sqrt{\frac{3n}{n+2}}$ for $p \in \mathbb{T}$. Then $\mathbb{T}$ is a Gromov torus.

Theorem 2. Gromov tori are flat, embedded in a sphere and minimal in it, and also $k_p(v) \equiv \sqrt{\frac{3n}{n+2}}$.

Theorem 3. For any $n \geq 3$, there exists a non-equivariant Gromov torus $\mathbb{T}^n$.

Outline

1) Gromov tori are flat and minimal in a sphere, and also

$$k_p(v) \equiv \sqrt{\frac{3n}{n+2}}.$$

2) $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$ defines a Gromov torus with lattice $A \iff$

$$\operatorname{Im} f \subset \mathbb{R}^{q+1}, \Delta f + nf = 0, |f| \equiv 1, |D_u f| \equiv 1, |D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}}$$

$$(u \in \mathbb{S}^{n-1}), f(x+a) = f(x) \ (a \in A).$$

3.a) $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$ defines a Gromov torus with lattice $A \iff$

$f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$ for $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$ and the conditions of item 2 are satisfied in terms of $\{c_k\}$.

3.b) f defines an equivariant torus $\iff c_k \perp c_l$.

4) Construction of a counterexample using item 3.

Basic Concepts

$L^n \looparrowright \mathbb{R}^q$. $\Pi_x : T_x L \rightarrow N_x L$, $\Pi_x(v, v) := \gamma_v''(0)$, where γ_v is a geodesic, $\gamma_v(0) = x$, $\gamma_v'(0) = v$. Π is the **second fundamental form**.

$k_x(v) := \Pi(v, v)$ is the **normal curvature**.

$H(x) := \sum_{i=1}^n k_x(e_i)$ is the **mean curvature**.

A submanifold is **minimal** if $H(x) \equiv 0$.

$\text{Sc}(x) := \sum_{i \neq j} \text{Sec}(e_i, e_j)$ is the **scalar curvature**.

$$\mathcal{K}(x) := \frac{1}{|\mathbb{S}^{n-1}|} \int_{|v|=1} |\Pi_x(v, v)|^2 dv.$$

Proof of Theorem 2

Item 1)

Theorem (Gauss Formula). $Sc = \frac{3}{2}|H|^2 - \frac{n \cdot (n+2)}{2} \cdot \mathbb{K}$

Theorem. g is a Riemannian metric on $\mathbb{T}^n$, $n \geq 3$, Sc is its scalar curvature, $u \in C^\infty(\mathbb{T}, \mathbb{R}_+)$. Then $(Sc \cdot u - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u) \cdot u^{\frac{n-2}{n+2}}$ is the scalar curvature for the metric $u^{\frac{4}{n-2}} \cdot g$.

Theorem. $\mathbb{T}^n$ is a torus with a Riemannian metric.

- 1) $\exists x : Sc(x) \leq 0$.
- 2) $Sc(x) \geq 0$ for $x \in \mathbb{T} \implies \mathbb{T}$ is flat.

Proof of Theorem 2

Using the theorems above, A. Petrunin proves the following inequalities in his paper:

Theorem. Let L be compact.

- 1) $L \looparrowright \mathbb{B}^q$. Then $\frac{1}{|L|} \int_L |H(x)| dx \geq n$.
- 2) $L \looparrowright \mathbb{S}^q$. Then $|H(x)| \geq n$ for $x \in L$.

Theorem. g is a Riemannian metric on $\mathbb{T}^n$, $n \geq 3$, Sc is its scalar curvature, $u \in C^\infty(\mathbb{T}, \mathbb{R}_+)$. Then $\exists x \in \mathbb{T}$:
 $Sc(x) \cdot u(x) - 4 \cdot \frac{n-1}{n-2} \cdot \Delta u(x) \leq 0$.

Proof of Theorem 2

We investigate the case of equality and obtain:

Lemma 1. $N^m \looparrowright \mathbb{S}^{n-1} \subset \mathbb{R}^n$. Then $|H_{N \subset \mathbb{R}^n}(p)| = m \implies |H_{N \subset \mathbb{S}^{n-1}}(p)| = 0$.

Lemma 2. $f : \mathbb{R}^n \rightarrow \mathbb{S}^q$ is a minimal isometric immersion. Then $\Delta f + nf = 0$.

As well as the proof of Theorem 2.

Defining a Gromov Torus

Item 2)

Let $\mathbb{T}^n \hookrightarrow \mathbb{S}^q \subset \mathbb{R}^{q+1}$ be a Gromov torus. $\mathbb{T}^n \cong \mathbb{R}^n / A\mathbb{Z}^n$. Thus, a Gromov torus is defined by a map $f : \mathbb{R}^n \rightarrow \mathbb{R}^{q+1}$ with the following properties:

- 1) $|f| \equiv 1$,
- 2) $|D_u f| \equiv 1 \ (u \in \mathbb{S}^{n-1})$,
- 3) $|D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}} \ (u \in \mathbb{S}^{n-1})$,
- 4) $f(x+a) = f(x) \ (a \in A)$,
- 5) $\Delta f + nf = 0$.

In fact, these requirements on f are also sufficient:

Proposition 1. $f : \mathbb{R}^n \rightarrow \mathbb{R}^{q+1}$ defines a Gromov torus with lattice $A \iff$ conditions 1) – 5) are satisfied.

Defining a Gromov Torus

Item 3.a)

For further work, we reformulate our criterion in a more convenient form.

Lemma 3. $f(x+a) = f(x)$ ($a \in A$), $\Delta f + nf = 0 \iff$
 $f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$ for $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$.

A map f defining a Gromov torus satisfies the conditions of Lemma 3. Then, from Proposition 1 and Lemma 3:

Proposition 2. $f : \mathbb{R}^n \rightarrow \mathbb{C}^{q+1}$ defines a Gromov torus with lattice $A \iff f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$ for $k \in \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*$,
 $\text{Im} f \subset \mathbb{R}^{q+1}$, $|f| \equiv 1$, $|D_u f| \equiv 1$, $|D_u^2 f| \equiv \sqrt{\frac{3n}{n+2}}$ ($u \in \mathbb{S}^{n-1}$).

Criterion in Terms of Coefficients

1.

$$\sum_{j \in M} |c_j|^2 = 1;$$

2.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha)^2 = \frac{1}{4\pi^2};$$

3.

$$\sum_{j \in M} |c_j|^2 k_j^\alpha k_j^\beta = 0, \alpha \neq \beta;$$

4.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha)^4 = \frac{3n}{16\pi^4(n+2)};$$

5.

$$\sum_{j \in M} |c_j|^2 (k_j^\alpha + k_j^\beta)^4 = \frac{3n}{4\pi^4(n+2)}.$$

Criterion in Terms of Coefficients

$$L_v := \{(i, j) \in M \times M \mid k_i - k_j = v\}, v \in \mathbb{R}^n, 2l_v := |L_v|.$$

1.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle = 0;$$

2.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle k_i^\alpha k_j^\beta = 0;$$

3.

$$\sum_{(i,j) \in L_v} \langle c_i, c_j \rangle (k_i^\alpha + k_i^\beta)^2 (k_j^\alpha + k_j^\beta)^2 = 0;$$

4.

$$\langle c_i, c_j \rangle = \overline{\langle c_j, c_i \rangle};$$

5.

$$\langle c_i, c_j \rangle = \langle c_{-j}, c_{-i} \rangle.$$

Equivariance Criterion

Item 3.b)

If we want to define a non-equivariant Gromov torus by a map f , we will need an equivariance criterion in terms of $\{c_k\}$.

Lemma 4. Let $f(x) = \sum_k c_k \cdot \exp(2\pi i \langle k, x \rangle)$ define a Gromov torus. Then the torus is equivariant $\iff c_k \perp c_l$ for $k \neq l$.

Proof Outline.

- 1) Torus is equivariant $\Rightarrow$ torus is an orbit of a torus in $O(q)$.
- 2) Torus is an orbit of a torus in $O(q) \Rightarrow$ Torus is equivariant.
- 3) Torus is an orbit of a torus in $O(q) \Rightarrow c_k \perp c_l$ for $k \neq l$.
- 4) $c_k \perp c_l$ for $k \neq l \Rightarrow$ torus is an orbit of a torus in $O(q)$.

A toral subgroup of a Lie group is any connected, compact, Abelian Lie subgroup.

Construction Outline

Item 4)

Step 1) Consider an equivariant Gromov torus. We add variables to its inhomogeneous system for $\{|c_j|^2\}$ so that the system has maximal rank and a positive solution.

Step 2) We straighten the obtained set of vectors while preserving the positive solution and complete it to a square lattice, so that some homogeneous system has a non-trivial solution.

Step 3) Using the obtained inner products, we find the coefficients $\{c_j\}$ themselves.

Notation

Step 1)

Let $\mathbb{T}^n$ be an equivariant Gromov torus with lattice A ,

$$\{k_j\}_{j=1}^m := \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1} \cap A^*.$$

$$\Phi(k) := (1, \dots, (k^j)^2, \dots, k^j \cdot k^i, \dots, (k^j)^4, \dots, (k^j + k^i)^4, \dots).$$

$$b := \left(\frac{1}{2}, \dots, \frac{1}{8\pi^2}, \dots, 0, \dots, \frac{3n}{32\pi^4(n+2)}, \dots, \frac{3n}{8\pi^4(n+2)}, \dots \right).$$

Then the inhomogeneous system with respect to $\{|c_j|^2\}$,
 $\sum_{j=1}^m \Phi(k_j) \cdot |c_j|^2 = b$, has a non-negative solution.

$$\{k_j\}_{j=1}^r := \{k_j : |c_j|^2 > 0\}.$$

Construction

$$V := \text{Span}(\text{Im} \Phi|_{\frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}}), \quad V_0 := \text{Span}(\Phi(k_1), \dots, \Phi(k_r)).$$

Lemma 5. $V \neq V_0$, let $\lambda_{\mathbb{S}}^{n-1}$ be the Lebesgue measure on $\frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}$, and μ be the left-invariant Haar measure on $O(n)$.

1) $\lambda_{\mathbb{S}}^{n-1}(\Phi^{-1}(V_0)) = 0$.

2) $\mu(\{\varphi \in O(n) : \Phi(\varphi(k_j)) \notin V_0, j = 1, \dots, m\}) = \mu(O(n))$.

Then, by Lemma 5, we can complete the lattice to the required one:

Corollary. $\exists \{\kappa_i\}_{i=1}^{\rho} \subset \frac{\sqrt{n}}{2\pi} \mathbb{S}^{n-1}$: $V = V_0 + \text{Span}(\Phi(\kappa_1), \dots, \Phi(\kappa_{\rho}))$ and $\sum_{j=1}^r \Phi(k_j) \cdot |c_j|^2 + \sum_{i=1}^{\rho} \Phi(\kappa_i) \cdot |c_i|^2 = b$ has a positive solution with respect to $\{|c_j|^2\}$.

Construction

Step 2)

B is a **square** lattice if its basis is proportional to an orthonormal one.

Using the corollary, we can perform the second step in the following sense:

Lemma 6. $\varepsilon_A > 0$. Then $\exists B$, an arbitrarily small square lattice, with $\text{dist}(k_j, \frac{\sqrt{n}}{2\pi} S^{n-1} \cap B) < \varepsilon_A$ ($j = 1, \dots, r$):

- 1) B^* defines a Gromov torus with non-zero coefficients $\{c_j\}$.
- 2) $\exists \{a_j\}, \{b_{ji}\} \subset \mathbb{C}$ is a solution to the entire system for the lattice B^* , $\forall a_j > 0, \exists b_{ji} \neq 0$.

Construction

Step 3)

Due to the homogeneity of the subsystems of interest, it suffices to prove the following lemma:

Lemma 7. Let $\{a_j\}, \{b_{ji}\} \subset \mathbb{R}, \forall a_j > 0$. Then
 $\exists N \in \mathbb{N}, \varepsilon > 0, \{v_j\} \subset \mathbb{R}^N: |v_j| = a_j, \langle v_j, v_i \rangle = \varepsilon b_{ji}.$

Lemmas 6.2 and 7 together complete our construction and prove Theorem 3.